SQL Assessment Report

E-commerce Sales Order Analysis | Assessment of 12 SQL queries, results, and findings.

1. Executive Summary

This assessment reviews SQL outputs for sales orders to identify patterns in order fulfilment, revenue, promotions, products, cancellations and returns, customer spending, and marketing channels. The output indicates that the largest concern is revenue loss from cancelled and returned orders, alongside substantial month-to-month revenue volatility.

Product and referral-source revenue are relatively distributed rather than dominated by one product or channel.

Orders analysed	Revenue in analysed rows	Average order value	Lost revenue share
1200	1,264,761.96	1,053.97	41.1%

2. Query-by-Query Assessment

Query 1 — Completed sales (Delivered / Shipped)

Problem: Management needs to understand revenue from orders that have reached the Delivered or Shipped statuses. The total orders and total revenue with Delivered and Shipped order statuses are compared.

	OrderStatus	total_order	total_revenue
1	Shipped	235	246159.58
2	Delivered	231	242600.32

SQL Query result showing Completed Sales analysis

Finding: Delivered orders total 231 and revenue of 242,600.32; Shipped orders total 235 and revenue of 246,159.58. Together, these statuses account for 466 orders and 488,759.90 in revenue—about 38.8% of the analysed orders. The remaining order statuses with 61.2% have not reached Delivered or Shipped.

Query 2 — High-value Pending orders

Problem: High-value orders left in Pending status may be at risk of being delayed or lost. Listing pending orders worth more than 2,000 for the 10 largest orders.

	OrderID	Product	Quantity	ReferralSource	TotalPrice
1	ORD201031	Phone	5	Email	3322.55
2	ORD200367	Laptop	5	Instagram	3293.85
3	ORD200540	Laptop	5	Referral	3243.25
4	ORD200221	Tablet	5	Google	3196.85
5	ORD200097	Tablet	5	Email	3097.6
6	ORD200241	Chair	5	Email	3078.35
7	ORD200876	Chair	5	Google	3044.55
8	ORD200816	Tablet	5	Referral	2873.95
9	ORD201002	Tablet	4	Referral	2799.52
10	ORD200628	Printer	5	Referral	2793.15

SQL Query Result showing pending orders at risk

Finding: The 10 largest Pending orders above 2,000 are mostly purchases of four or five units. The largest is a Phone order worth 3,322.55 attributed to Email; the 10 listed orders are worth more than 30,743.62 combined. Prioritising these orders for customer or payment follow-up and track whether they are fulfilled or cancelled could protect meaning revenue.

Query 3 — Coupon performance

Problem: The business needs to understand the order volume and revenue associated with coupon campaigns. Counting orders and revenue for each coupon code.

	CouponCode	total_order	total_revenue
1	Freeship	313	335036.99
2	Save10	286	304840.02
3	Winter15	292	302483.54

SQL Query Result showing Coupon Code Performance

Finding: Freeship leads with 313 orders and 335036.99 revenue, Winter15 with 292 (302,483.54) and Save10 (304,840.02). Average revenue per order is broadly similar across the coupons and No code coupon is also profitable. Freeship has the greatest volume, but profitability cannot be determined from revenue alone; compare discount cost, margin, and conversion before selecting a winning offer.

Query 4 — Revenue and cart size by product

Problem: Identify the products generating the most revenue and compare average cart size.

	Product	total_revenue	total_order	avg_items_in_cart
1	Chair	195620.11	178	5.78089887640449
2	Printer	195612.61	181	5.51381215469613
3	Laptop	192126.56	173	5.34104046242775
4	Tablet	186568.95	179	5.50279329608939
5	Monitor	175651.41	163	5.55828220858896
6	Desk	167459.93	170	5.57647058823529
7	Phone	151722.39	156	5.07692307692308

SQL Query Result showing Product Revenue and Order volume

Finding: Chair leads with 195,620.11 revenue, closely followed by Printer at 195,612.61. Phone is lowest at 151,722.39. Average items in cart range from 5.1 to 5.8. Investigating product price, and order volume for Phone; the similar cart-size averages do not by themselves explain revenue differences.

Query 5 — Cancellation rate by payment method

Problem: Determine whether cancellation rates vary by payment method.

	PaymentMethod	total_orders	cancelled	cancelled_rate_pct
1	Credit Card	234	54	23.08
2	Gift Card	230	50	21.74
3	Online	258	53	20.54
4	Cash	246	49	19.92
5	Debit Card	232	44	18.97

SQL Query Result showing Cancellation Rate by Payment Method

Finding: Credit Card has the highest cancellation rate at 23.1%, closely followed by Gift Card at 21.8%. Debit card has the lowest rate at 19.0%; the difference between the highest and lowest is 4.1 percentage points. Investigating cancellation reasons by payment method before concluding that payment friction causes the difference.

Query 6 — Revenue by referral source

Problem: Identify which marketing/referral sources contribute the most revenue, considering sources with more than 50 orders.

	ReferralSource	total_orders	total_revenue
1	Instagram	259	275285.45
2	Email	250	261808.55
3	Google	241	250441.48
4	Facebook	228	250410.9
5	Referral	222	226815.58

SQL Query Result Showing Referral source performance

Finding: Instagram generates 275,285.45 across 259 orders, followed closely by Email at 261,88.55 (250), Google at 250,441.48 (241), and Facebook at 250,410.9 (228). Referral is lower at 226,815.58 across 222 orders. Performance among the referral sources is close and business is not over reliant on one source. Compare campaign spend and acquisition cost, not revenue alone, to asses performance.

Query 7 — Lost versus retained revenue

Problem: Quantify revenue associated with cancelled and returned orders compared with all other statuses.

retained_revenue	loss_revenue	total_revenue	retained_pct	loss_pct
745088.05	519673.91	1264761.96	58.91	41.09

SQL Query Result Showing Percentage of Lost and Retained Revenue

Finding: Cancelled and Returned orders are grouped as Lost, totalling 519,673.93 (41.1%). All other statuses are grouped as Retained, totalling 745,088.05 (58.9%). This is the clearest improvement opportunity in the output. However, the query defines Retained as every status other than Cancelled or Returned, so it include Pending orders; it should not be interpreted as confirmed collected revenue.

Query 8 —Top 10 customer by total spend

Problem: Identify Customers who made the biggest purchases. h. Using CTE to aggregate total spend per customer.

	CustomerID	Quantity	total_orders	total_spent
1	C38840	4	2	5723.23
2	C57276	5	1	3456.4
3	C67260	5	1	3390.8
4	C13877	5	1	3384.9
5	C18404	5	1	3370.2
6	C16775	5	1	3353.75
7	C65986	5	1	3352.4
8	C47778	5	1	3334
9	C59183	5	1	3322.55
10	C25276	5	1	3313.9

Finding: C38840 has the highest spend with 5,723.23 and 2 orders, the other nine customers spent largely on single order The top 10 spenders spent 36,002.13 altogether, which has 2.8% of total revenue.

Query 9 — Product revenue ranking and share

Problem: Rank products by revenue and determine each product's share of total revenue.

	Product	total_revenue	revenue_rank	pct_of_total_revenue
1	Chair	195620.11	1	15.47
2	Printer	195612.61	2	15.47
3	Laptop	192126.56	3	15.19
4	Tablet	186568.95	4	14.75
5	Monitor	175651.41	5	13.89
6	Desk	167459.93	6	13.24
7	Phone	151722.39	7	12.0

SQL Query Results Showing Product Ranking

Finding: Chair ranks first and Printer second, each contributing 15.5% of revenue. Phone ranks seventh at 12.0%. No product contributes more than 16% of total revenue. Revenue is spread across the product range and the business is not exposed to failure of one product. Using the rank to guide merchandising while monitoring margins and stock availability.

Query 10 — Average order value: coupon versus no coupon

Problem: Check whether orders using coupons have a different average order value from orders without coupons.

	Segment	total_orders	avg_order_value
1	Coupon used	891	1057.64
2	No coupon	309	1043.37

SQL Query Result Showing Average Order Value Across Coupon used and No coupon

Finding: Coupon orders have an average order value of 1,057.64 across 891 orders, compared with 1043.37 across 309 orders without a coupon—about 1.4% higher. Coupon use is associated with a higher average order value in this dataset, but the comparison does not prove coupons caused the increase. Compare profit after discounts and account for customer/order differences.

Query 11 — Monthly completed-sale revenue and growth

Problem: Assess whether monthly revenue from Delivered and Shipped orders is stable or growing.

	month	revenue	num_orders	month_over_month_change	mom_growth_pct
1	2023-01	13339.68	11	NULL	NULL
2	2023-02	8396.83	9	-4942.85	-37.1
3	2023-03	16099.56	12	7702.73	91.7
4	2023-04	13776.88	14	-2322.68	-14.4
5	2023-05	28673.9	20	14897.02	108.1
6	2023-06	21098.47	19	-7575.43	-26.4
7	2023-07	24352.3	18	3253.83	15.4
8	2023-08	18145.2	18	-6207.1	-25.5

9	2023-09	5640.58	6	-12504.62	-68.9
10	2023-10	36899.08	28	31258.5	554.2
11	2023-11	17831.63	17	-19067.45	-51.7
12	2023-12	11424.24	12	-6407.39	-35.9
13	2024-01	18058.81	16	6634.57	58.1
14	2024-02	10199.14	12	-7859.67	-43.5
15	2024-03	19015.83	19	8816.69	86.4
16	2024-04	15864.21	15	-3151.62	-16.6
17	2024-05	13080.22	14	-2783.99	-17.5
18	2024-06	22831.88	24	9751.66	74.6
19	2024-07	16281.54	15	-6550.34	-28.7
20	2024-08	7533.54	8	-8748.0	-53.7
21	2024-09	17641.81	21	10108.27	134.2
22	2024-10	20036.04	15	2394.23	13.6
23	2024-11	16694.19	18	-3341.85	-16.7
24	2024-12	22141.39	20	5447.2	32.6
25	2025-01	7270.75	9	-14870.64	-67.2
26	2025-02	10045.18	15	2774.43	38.2
27	2025-03	15527.5	22	5482.32	54.6
28	2025-04	6543.05	8	-8984.45	-57.9
29	2025-05	15045.26	13	8502.21	129.9
30	2025-06	19271.21	18	4225.95	28.1

SQL Query Result Showing Month Over Month Growth

Finding: Monthly revenue ranges from 5,640.58 in September 2023 to 36,899.08 in October 2023. +554,2% of Month-over-month growth varies sharply. Growth moves between large rises and falls. Revenue is volatile and there is no consistent upward trend in these monthly results. Check seasonality, promotions, order mix, and unusually high/low months before forecasting.

Query 12 — Referral-source funnel

Problem: Determine which referral sources produce the highest share of Delivered/Shipped orders and their average order value.

	ReferralSource	total_orders	success_rct_pct	avg_order_value
1	Referral	222	41.4	1021.69
2	Facebook	228	39.9	1098.29
3	Google	241	38.6	1039.18
4	Email	250	38.4	1047.23
5	Instagram	259	36.3	1062.88

Finding: Referral has the highest success rate at 41.4%, while Facebook has the highest average order value at 1,098.29. Instagram has the lowest success rate at 36.3%. The success-rate gap is approximately 5.1 percentage points. Facebook is a promising channel to investigate further, while Instagram may need a closer review of the path from click to completed order.

3. Overall Findings

Priority	Assessment
1. Revenue leakage	Cancelled and Returned orders represent 41.1% of revenue under the report's definition. Validate status classification and focus on reasons for cancellations and returns.
2. Order fulfilment	Only 466 of 1200 orders are Delivered and Shipped. Investigate the other statuses and ageing of Pending orders.
3. Revenue volatility	Monthly completed-sale revenue changes substantially from month to month, making trend monitoring and forecasting important.
4. Commercial performance	Chair and Printer lead product revenue; Google, Email, Instagram, and Facebook produce similar total revenue.
5. Promotions	Coupon orders show higher average order value.

4. Recommendations

- Prioritise a root-cause review of cancelled and returned orders; separate pending, shipped, delivered, cancelled, and returned revenue in reporting.
- Contact customers or resolve operational/payment issues for the highest-value Pending orders first.
- Track monthly fulfilled revenue alongside order count, cancellation rate, return rate, and average order value.
- Assess marketing channels using conversion rate, acquisition cost, and profit—not revenue alone.
- Evaluate coupon campaigns using net revenue and margin after discounts; test whether the higher average order value translates into higher profit.

5. SQL and Data Quality Notes

- **Data Note:** The output includes 1,200 orders from the cleaned data sheet that included ORD200000- ORD201199.
- **Method:** DB Browser SQLite was used to run SQL queries and output.
- **Interpretation:** SQL outputs show patterns and associations. They do not, on their own, prove why cancellations occur or that coupons/marketing channels cause better outcomes.

Conclusion: The SQL assessment highlights revenue leakage and inconsistent fulfilment as the most urgent areas for investigation. The next step is to validate status definitions, investigate cancellation/return causes, and rerun the analysis on the complete dataset before finalizing business decisions.